

Patient Priority of Proposed U.S. Bills for Physical AI Oncology Clinical Trials

Proposed Legislation for Improving Future
Cancer Patients' Control over their Disease

May 7, 2026

Kevin Kawchak

CEO ChemicalQDevice

<https://orcid.org/0009-0007-5457-8667>

<https://doi.org/10.5281/zenodo.20045457>

Unofficial independent draft derived from prior legislation; no third-party endorsement, sponsorship, affiliation, or authorization is expressed or implied; and was adapted using Claude Code Opus 4.7 and ChatGPT 5.5 Thinking.

This work was adapted from original CFR documents in the public domain. The original ICH document is copyrighted and may be used, reproduced, incorporated into other works, adapted, modified, translated or distributed under a public license. This current work is not endorsed or sponsored by CFR, ICH, or FDA; and was adapted using Claude Code.

This draft is independent and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, medical society, or technology vendor. Bill numbers and short titles are illustrative placeholders selected to avoid known active legislation under the 119th Congress; the legal-design intent of each section remains the substantive contribution.

Table of Contents

1	HR 9501 (2026) - Cancer Patient Self-Selection of Physical AI Oncology Trials Act of 2026 (Adaption of 21 CFR Part 50, FDA Decentralized Trial Final Guidance 2024, and 42 USC 300gg-8)	4
1.1	Findings and Declarations	4
1.2	Definitions	5
1.3	Patient Self-Selection Rights	6
1.4	Implementation	6

1.5	Reporting and Enforcement	7
1.6	Patient-Control Framings	7
2	HR 9502 (2026) - Cancer Patient Robot-and-Humanoid Choice Act of 2026 (Revision of California AB 2847 (2026) and FDA AI Regulatory Decision-Making Draft Guidance 2025)	8
2.1	Findings and Declarations	8
2.2	Definitions	9
2.3	Patient Robot Choice Rights	9
2.4	Implementation	10
2.5	Reporting and Enforcement	10
2.6	Discussion and Limitation	11
2.7	Patient-Control Framings	11
3	HR 9503 (2026) - Cancer Patient Procedural Modification Authority Act of 2026 (Adaption of OHRP Broad Consent 2017 and 21st Century Cures Act 2016)	12
3.1	Findings and Declarations	12
3.2	Definitions	12
3.3	Patient Procedural Modification Rights	13
3.4	Implementation	13
3.5	Reporting and Enforcement	14
3.6	Discussion and Limitation	14
3.7	Patient-Control Framings	14
4	HR 9504 (2026) - Physical AI Clinical Error Reduction Act of 2026 (Adaption of HTI-1 DSI Final Rule 2023 and FDA AI Decision-Making Draft Guidance 2025)	15
4.1	Findings and Declarations	15
4.2	Definitions	16
4.3	Error Rate Reduction Rights	16
4.4	Comparison Diagram (Human Baseline vs Physical AI)	16
4.5	Implementation	17
4.6	Reporting and Enforcement	17
4.7	Implementation Guardrails	18
4.8	Discussion and Limitation	18
4.9	Patient-Control Framings	18
5	HR 9505 (2026) - Real-Time Patient-Sponsor Direct Communication Act of 2026 (Revision of FDA Real-Time Clinical Trials Press Announcement April 2026 and FDA Decentralized Clinical Trials Guidance 2024)	19
5.1	Findings and Declarations	19
5.2	Definitions	19
5.3	Patient-Sponsor Direct Communication Rights	20
5.4	Implementation	20
5.5	Reporting and Enforcement	21
5.6	Discussion and Limitation	21

5.7	Patient-Control Framings	21
6	HR 9506 (2026) - American Physical AI Oncology Leadership Act of 2026 (New Federal Authorizing Statute Adapting FDORA Section 3209 and 21st Century Cures Act 2016)	22
6.1	Cross-Bill Implementation Timeline	22
6.2	Findings and Declarations	23
6.3	Definitions	23
6.4	American Leadership Authorizations	24
6.5	New Physical-AI Reversal Guardrail	24
6.6	Implementation	24
6.7	Reporting and Enforcement	24
6.8	Future Work Tracks	25
6.9	Future Work Deliverables	25
6.10	Persistent Themes Across the Seven Bills	25
6.11	Implications for Cancer Patient Control	26
6.12	Discussion and Limitation	26
6.13	Forward Path	26
6.14	Patient-Control Framings	26
7	HR 9507 (2026) - Cancer Patient Health Data Self-Custody and Trial-Selection Act of 2026 (Revision of HHS HIPAA Right-to-Access 2025 and ONC Cures Act Final Rule)	28
7.1	Findings and Declarations	28
7.2	Definitions	28
7.3	Patient Health Data Self-Custody Rights	29
7.4	Implementation	29
7.5	Reporting and Enforcement	29
7.6	Discussion and Limitation	30
7.7	Patient-Control Framings	30
	References	35
	Acknowledgments	36
	Ethical disclosures	36
	Rights and permissions	36
	Cite this article	36

1. HR 9501 (2026) - Cancer Patient Self-Selection of Physical AI Oncology Trials Act of 2026 (Adaption of 21 CFR Part 50, FDA Decentralized Trial Final Guidance 2024, and 42 USC 300gg-8)

The cancer patient, not the doctor, the nurse, the trial sponsor, the IRB, or the regulator, is the priority participant in any United States oncology clinical trial. Advanced AI and advanced robotics now offer the patient direct control over discovery, enrollment, scheduling, robot selection, procedural modification, real-time data feedback, and self-custody of health data. HR 9501 is the foundational bill of seven proposed federal bills (HR 9501 through HR 9507) that supply the legal infrastructure for that control. The bill is an *Adaption of 21 CFR Part 50, FDA Decentralized Clinical Trial Final Guidance 2024, and 42 USC 300gg-8*, drawing the chain of prior authority from the Common Rule baseline through the FDA decentralized elements guidance and the Medicaid coverage statute that already binds qualifying clinical trials [1, 2, 3, 4, 5, 6, 7].

The current legislative stack supplies the points of departure for this bill. The CLINICAL TREATMENT Act of 2020 (PL 116-260) established Medicaid coverage of routine costs in qualifying trials from January 1, 2022; HR 9501 extends that coverage right into a positive self-selection right that includes 24/7 booking, decentralized payment parity, and AI trial-matching tool actionability for Medicaid-eligible patients [8, 9, 10, 11]. Affordable Care Act Section 2709 (PL 111-148, 2010) set the private-insurance baseline for routine clinical-trial cost coverage; HR 9501 closes the decentralized and at-home component payment-parity gap that ACA Section 2709 left unaddressed [12, 4]. FDAAA Section 801 (PL 110-85, 2007) established the ClinicalTrials.gov infrastructure; HR 9501, jointly with HR 9507, closes the structured-eligibility-data gap that prevents patient-initiated trial discovery [13]. HHS HIPAA Right-to-Access (2025), the ASTP/ONC Patient Portals Brief 2024, and the ASTP/ONC Hospital APIs Brief 2024 establish the foundational data-access framework on which self-selection depends; the self- custody right itself is reserved for HR 9507 [14, 15, 16].

HR 9501 implements two of the seven dimensions of the operational definition of patient control that anchor this paper. The trial-discovery dimension (machine-readable eligibility, AI matching, no provider gatekeeping) is implemented through the Subsection 1.3 rights below; the location and scheduling dimension (24/7 booking, decentralized elements, home-based delivery) is implemented through the same operative core [9, 10, 11, 2, 3, 17, 5].

1.1. Findings and Declarations

Congress finds and declares the following.

(a) Approximately seven percent of adults with cancer in the United States enroll in an oncology clinical trial, despite documented patient-benefit advantages of trial participation [5]. The participation gap arises in part because the prior framework requires provider-initiated trial referral; patient-initiated discovery is technically possible but is not yet a statutory right [8, 12].

(b) Large language model-based trial-matching systems have demonstrated recall and accuracy at or above the clinician baseline. TrialGPT, a Nature Communications 2024 system, achieved a top-10

recall of 87.3 percent on a held-out test set; PRISM extended these numbers across multi-site EHR data; TrialMatchAI was published in Nature Communications 2026 with end-to-end recommendation latency on the order of seconds [9, 10, 11].

(c) The FDA Decentralized Clinical Trials Final Guidance 2024, the FDA Oncology Center of Excellence Advancing Oncology Decentralized Trials directory, and the FDA Real-Time Clinical Trials press announcement of April 28, 2026 together establish a regulatory opening for patient-initiated trial selection at sites that admit decentralized elements [2, 3, 18].

(d) The patient-side technical substrate is documented in the author’s prior 24/7 continuous trial paper at `new-trial/national-24-7-trial/paper/full-paper/`, in the single-patient 10-stage journey at `patient-journey/`, and in the autonomous sponsor 168-hour extension at `sponsor/final_paper/168_hours/`. These prior simulations demonstrate that hourly trial activity, daily AE escalation, and weekly cumulative reporting all run on commodity hardware [5, 6, 19].

(e) The CLINICAL TREATMENT Act of 2020 (PL 116-260) covers routine costs of qualifying trials for Medicaid beneficiaries from January 1, 2022, but does not establish a positive patient-initiated booking right [8].

(f) Affordable Care Act Section 2709 (PL 111-148, 2010) provides a private-insurance baseline for routine clinical-trial cost coverage but does not reach the decentralized-component payment-parity case that HR 9501 closes [12, 4].

(g) Mazor 2025 (JAMA Network Open) reported a null result on AI notifications alone in a randomized clinical trial of cancer progression alerts; this finding informs the bill by indicating that self-selection rights without a complete pathway from discovery to consent to enrollment may not improve participation on their own [20].

1.2. Definitions

For the purposes of this Act, the following definitions apply.

(a) “*Patient self-selection*” in this Act means the patient’s right to discover, evaluate, and book the patient’s own oncology clinical trial slot without intermediary provider gate-keeping. The current operational baseline is the existing 24/7 trial format documented in `new-trial/national-24-7-trial/paper/full-paper/main.tex` and the per-hour patient records files at `new-trial/national-24-7-trial/hour-00/` through `hour-55/` [5].

(b) “*Machine-readable eligibility report*” means a structured FHIR-based output containing the patient’s eligibility-relevant data fields for at least 80 percent of currently recruiting United States oncology trials. The structured-eligibility baseline is the ClinicalTrials.gov infrastructure under FDAAA Section 801 (PL 110-85, 2007) [13].

(c) “*Qualified Physical AI oncology trial site*” in this specific Act designates a site authorized under either in this context or under the prior California SB 1042 / Title 22 framework regulation documented at `new-trial/site/05-state-regulations/` and at `national-platform/new_trial_psl/05_title22_regulations.tex` [21, 22].

(d) “24/7 booking” means the patient’s right to book any qualified site at any hour of the day, any day of the week. The operational baseline is the 168-hour autonomous sponsor extension at `sponsor/final_paper/168_hours/` [19].

(e) “AI matching tool” means an algorithm that proposes trials given the patient’s machine-readable eligibility report. The qualified instances enumerated in this bill include TrialGPT, PRISM, and TrialMatchAI [9, 10, 11]; equivalent successors qualify under the HTI-1 DSI transparency framework [23].

1.3. Patient Self-Selection Rights

The patient holds the following rights under this Act, enforceable against any CMS-billable entity, qualified Physical AI oncology trial site, or sponsor.

(a) Right to receive a machine-readable eligibility report, in FHIR format, from any CMS-billable EHR within seven calendar days of pathology or genomic result release.

(b) Right to discover all currently recruiting United States oncology trials via an FDA-published API that integrates ClinicalTrials.gov, the FDA RTCT pilot directory, and the FDA Oncology Center of Excellence Advancing Oncology Decentralized Trials directory [3, 18].

(c) Right to book a 24/7 continuous trial patient slot at any qualified physical AI oncology clinical trial site without prior written referral. The operational pattern is documented in `new-trial/national-24-7-trial/hour-NN/diagram_facility.txt` across hours 00, 12, 23, 47, and beyond [5].

(d) Right to payment parity for decentralized and at-home trial components under 42 USC 300gg-8 and CMS NCD 310.1; this closes the gap left by ACA Section 2709 [4, 24].

(e) Right to cross-state portability of eligibility data and trial bookings, drawing on the technical baseline in the ASTP/ONC Hospital APIs Brief [16].

(f) Right to refuse provider-only routing of trial discovery in favor of an AI matching tool of the patient’s choosing (TrialGPT, PRISM, TrialMatchAI), with disclosure of model provenance under HTI-1 DSI transparency [23].

(g) Right to a single-click booking confirmation that creates an audit-trail record at the clinical trial sponsor’s pharmacovigilance endpoint, per the architecture documented at `sponsor/final_paper/scripts/core_agents/audit_trail_manager.py` [19].

1.4. Implementation

(a) The Secretary of Health and Human Services, acting through the Food and Drug Administration, the Office of the National Coordinator for Health Information Technology, and the Centers for Medicare and Medicaid Services, shall publish the unified federal API described in Subsection 1.3(b) within 12 months of the date of enactment.

(b) Each State, acting through its existing health-information infrastructure, shall achieve full self-selection-portal coverage across CMS-billable EHRs within 24 months of enactment.

(c) Implementation shall preserve stronger State-level patient protections, including the California SB 37 routine-cost coverage floor, under the preemption rule set forth in this Subsection [25, 26].

(d) The FDA, ONC, HHS, and CMS shall jointly publish, within 18 months of enactment, an implementation guidance document that references this Act and the FDA Decentralized Clinical Trials Final Guidance 2024 [2].

1.5. Reporting and Enforcement

(a) The Secretary shall publish, annually, a report on patient self-selection rates by State, by payer, and by cancer type. The federal Patient Control Dashboard’s Domain 1 (Access) metrics anchor this report: trial-offer rate, days from pathology or genomic result to trial shortlist, machine-readable eligibility report coverage, real-time data release rates, and travel and work hours saved [14, 27].

(b) Information-blocking penalties under [28] apply to any CMS-billable entity that fails to expose the self-selection portal endpoint required by Subsection 1.4.

(c) No patient shall be denied trial consideration by automated output alone. Each automated rejection triggers a human reviewer escalation under the HTI-1 DSI framework [29, 23].

1.6. Patient-Control Framings

For the individual patient, an NSCLC patient at home in rural Texas can, on the patient’s own initiative, receive an FHIR eligibility report within seven days of the patient’s genomic result, then self-book a 24/7 slot at a USL-7-or-above surgical robot system, then follow the 10-stage journey for PAT-2026-0042 documented at `patient-journey/stage_01_prescreening.py` through `stage_10_closeout.py` [6].

For broad adoption, all United States oncology trials must publish eligibility criteria in machine-readable form within the 12-month federal-API window, with all CMS-billable EHRs maintaining a self-selection portal endpoint accessible by every United States cancer patient by month 24. The United States must remain Number 1 in the world for cancer patient benefit during the medical AI and robotics revolution. HR 9501 is the foundational bill that opens that pathway, replacing the prior provider-gatekept trial referral pattern with patient-initiated, AI-matched, 24/7-bookable self-selection across all qualified Physical AI oncology trial sites [5, 6, 8].

2. HR 9502 (2026) - Cancer Patient Robot-and-Humanoid Choice Act of 2026 (Revision of California AB 2847 (2026) and FDA AI Regulatory Decision-Making Draft Guidance 2025)

HR 9502 is a recently proposed *Revision of California AB 2847 (2026) Patient Rights and Robotic Safety Act and FDA AI Regulatory Decision-Making Draft Guidance of 2025*, drawing the chain of prior authority from the California state-level patient-rights baseline up to the FDA capability-neutrality draft [30]. The California baseline is documented at `new-trial/site/02-legislation-patient-rights/physical_ai_patient_rights.tex` and at `national-platform/new_trial_psl/02_ab2847_patient_rights.tex` [21, 22]. HR 9502 implements the seventh dimension of the operational definition of patient control introduced in this set of proposed bills: physical-AI procedure execution choice. The patient selects which surgical robot, humanoid companion, or rehabilitation exoskeleton performs the patient’s care from the ten robot categories enumerated at `patients/patient_robot_instructions_fixed.tex` pages 1 through 10 and at `national-platform/patient_robot/01_preamble_robots1to5.tex` (surgical, cobot, RT positioning, needle placement, social companion) and `02_robots6to10_closing.tex` (humanoid, RT motion-tracking, imaging, steerable needle, rehabilitation exoskeleton) [31, 32, 33].

The operational definition of cancer patient control across six prior functional dimensions (clinical trial discovery, enrollment comprehension, location choice, data and biospecimen governance, toxicity warnings, AI-exclusion contestability) is supplied in the repository through `patients/paper/Deep-Research-3/part1_legal_baseline.md`. This paper introduces the seventh dimension that HR 9502 implements [1, 34, 2].

2.1. Findings and Declarations

Congress finds and declares the following.

(a) Ten distinct robot categories are clinically deployed in oncology care today: surgical robots, collaborative cobots, RT positioning robots, robotic needle-placement systems, social companion robots, humanoids, RT motion-management robots, imaging assistant robots, steerable needle robots, and rehabilitation exoskeletons [31].

(b) The Unification Standard Level (USL) framework in this bill provides a four-dimension scoring sheet (capability, safety, autonomy, trust) that quantifies each of the clinical trial robot’s readiness for patient choice on a 0 through 100 scale; the USL framework is documented at `national-platform/usl_standard/01_preamble_introduction_framework.tex` and `02_results_discussion_closing.tex` [32].

(c) Direct robotics evidence supporting patient-led robot choice includes the TARGET multicenter robotic-bronchoscopy trial published in CHEST 2025 (safety and diagnostic outcomes across multiple sites) and the NEJM Catalyst Cancer CARE Beyond Walls home-based chemotherapy study published in 2026 (safety, feasibility, and patient experience metrics) [33, 17].

(d) The California AB 2847 (2026) Patient Rights and Robotic Safety Act establishes patient rights at the State level only; HR 9502 extends those rights to national coverage through revision rather than displacement of the State framework [21].

(e) The FDA AI Regulatory Decision-Making Draft Guidance of January 2025 articulates the capability-neutrality principle for AI systems used in regulatory decision-making; HR 9502 incorporates that principle into its definitions of qualified robot [30].

(f) Foundation-model evidence in computational pathology (Virchow, Nature Medicine 2024) demonstrates that AI systems can reach clinical-grade performance across pan-cancer detection; the same pattern generalizes to robot-executed image-guided procedures [35].

2.2. Definitions

(a) “*Patient robot choice*” means the patient’s right to select which surgical robot, humanoid robot, or companion robot performs the patient’s care, exercised after review of the qualified robot comparison required by Subsection 2.3(a). The reference list of robot categories is the ten-category enumeration at `patients/patient_robot_instructions_fixed.tex` pages 1 through 10 [31].

(b) “*Qualified robot*” means a Physical AI system that meets the Unification Standard Level (USL) threshold for the system’s category, as documented at `national-platform/usl_standard/` and at `unification/usl/paper/usl_oncology_trials.tex` [32].

(c) Robot category in this particular bill means one of the ten patient-facing categories enumerated in Subsection 2 of the prior California SB 1042 baseline, repeated at `new-trial/site/01-legislation-authorization/physical_ai_trial_authorization.tex` and at `national-platform/new_trial_psl/01_preamble_and_sb1042.tex`, with each category carrying its own USL minimum threshold [21].

(d) “*Capability-neutral*” means the bill’s definitions apply to any robot architecture (LLM-driven, foundation-model, agentic system, or specialized hardware) that meets the USL threshold, following the second guardrail in `patients/paper/Deep-Research-3/part2_ai_robotics_legislation.md` and the FDA AI Regulatory Decision-Making Draft 2025 [30].

(e) “*USL transparency report*” means the four-dimension USL scoring sheet (capability, safety, autonomy, trust) for a specific robot instance, published in machine-readable form by the qualified site that operates the robot [32].

2.3. Patient Robot Choice Rights

(a) Right to receive a side-by-side comparison of all qualified robots for the patient’s cancer procedure, including USL scores, prior outcomes, and named instances. Worked examples drawn from the single-patient journey include da Vinci Xi at USL 87.5 for lobectomy (`patient-journey/stage_05_surgery.py`), post-operative cobot for limb stabilization (`patient-journey/stage_06_recovery.py`), Franka Emika at USL 88.75 for pembrolizumab infusion (`patient-journey/stage_07_immunotherapy.py`), and a companion robot for pediatric or geriatric surveillance (`patient-journey/stage_09_surveillance.py`) [6].

(b) Right to select a specific robot instance (for example RTPOS-01 versus RTPOS-03) following the per-hour `patient_records.md` naming pattern located at repository file identification `new-trial/national-24-7-trial/hour-12/patient_records.md` [5].

(c) Right to substitute a humanoid robot (Tesla Optimus, Boston Dynamics Atlas, Agility Digit, or other USL-qualified humanoid evaluated under `unification/usl/humanoids/`) for routine non-surgical tasks (transport, vitals collection, medication delivery) where the patient prefers reduced human contact [32].

(d) Right to refuse a specific robot instance and request a substitute at no added patient cost.

(e) Right to receive a USL transparency report for any robot before consenting to the use [32].

(f) Right of appeal to the FDA Oncology Center of Excellence DCT pathway for any robot denial that the patient believes is capability-equivalent to the requested robot [3].

(g) Right to a humanoid-robot deployment for in-home care over the course of long therapeutic regimens (the 35 pembrolizumab cycles for PAT-2026-0042 are a worked example), with a patient-side override authority that may revert any in-home humanoid deployment to human- nurse delivery within the same calendar day [6].

2.4. Implementation

(a) The Secretary of Health and Human Services, acting through the Food and Drug Administration, the Agency for Healthcare Research and Quality, and the Office of the National Coordinator for Health Information Technology, shall require all qualified Physical AI oncology trial sites to publish a per-robot USL transparency report within 12 months of enactment.

(b) The bill is a Revision of California AB 2847 (2026) for national coverage; State-level protections that exceed the federal floor are preserved by the preemption rule [21].

(c) Implementation references the operational pattern at the repository file identification `national-platform/patient_robot/02_robots6to10_closing.tex` and the safety-workflow integration at `sponsor/final_paper/scripts/safety/` [22, 19].

2.5. Reporting and Enforcement

(a) Annual federal report listing the per-state, per-cancer, per-robot-category selection rate. Domain 5 of the federal Patient Control Dashboard (Physical-AI Execution) anchors this report: patient-selected robot percentage by category, patient-initiated procedural modification rate (tied to HR 9503), patient-sponsor direct channel uptake (tied to HR 9505), and the American Leadership Index ranking (tied to HR 9506) [31, 32, 22].

(b) The AI-decision-tracking infrastructure at `sponsor/final_paper/scripts/core_agents/` (53 agents listed in `sponsor/final_paper/scripts/core_agents/__init__.py`) provides the FDA-auditable substrate [19].

(c) Civil penalties harmonize with the FDA AI Regulatory Decision-Making Draft 2025 capability-neutrality enforcement framework [30].

2.6. Discussion and Limitation

The bill rests on the assumption that USL scoring is comparable across vendors. The unification/usl/ paper documents that USL scoring requires multi-site calibration; humanoid robots in particular have a thinner clinical evidence base than surgical robots. The findings above acknowledge this limitation and constrain humanoid deployment to non-surgical routine tasks until additional prospective evidence accumulates [32].

2.7. Patient-Control Framings

For the individual patient, a 58-year-old female NSCLC patient matching the PAT-2026-0042 profile chooses the da Vinci Xi for her lobectomy after reviewing USL 87.5 versus the USL 84.0 alternative, and chooses a Tesla Optimus humanoid for home-based vitals collection during her 35 pembrolizumab cycles, with a daily revert-to-human- nurse override available at any point during the regimen [17, 6].

For broad adoption, all Physical AI oncology trial sites must publish a USL transparency report for every deployed robot instance within 12 months. The bill positions the United States as Number 1 in the world for patient-controlled robot selection, replacing the prior provider-only choice pattern under traditional CA AB 2847 with a national patient-led standard governing the ten robot categories [31, 21].

3. HR 9503 (2026) - Cancer Patient Procedural Modification Authority Act of 2026 (Adaption of OHRP Broad Consent 2017 and 21st Century Cures Act 2016)

HR 9503 is an *Adaption of OHRP Broad Consent Recommendations 2017 and the 21st Century Cures Act of 2016 (PL 114-255)*, drawing the chain of prior authority from the Common Rule baseline at 45 CFR 46 through the federal broad-consent recommendations and the Cures Act information-blocking statute that already binds interoperable health information technology [36, 37, 1]. The bill implements the second dimension of the operational definition of patient control: enrollment comprehension, key information rules, broad consent, and patient-initiated translation of plain-language summaries with real-time consent updates [34, 38].

The current legal stack supplies the foundation for HR 9503. FDA Informed Consent Guidance 2023, the FDA / OHRP Key Information Draft of 2024, and the OHRP Broad Consent Recommendations of 2017 establish the consent and broad-consent frameworks. HR 9503 closes the patient-control gap by adding a positive procedural-modification right where the patient may modify the patient’s own treatment under physical-AI execution, governed by a real-time consent update logged within five minutes of the modification [34, 38, 36].

3.1. Findings and Declarations

(a) The current consent framework permits broad consent for secondary biospecimen use under the Common Rule, but procedure-level consent for the index trial remains binary: accept the entire protocol or refuse [1, 36].

(b) Physical AI executors with USL scores at or above the modification-tier threshold can perform per-cycle and per-step patient-initiated modifications with no documented loss of patient safety. The 35 pembrolizumab cycles managed by Franka Emika at USL 88.75 in `patient-journey/stage_07_immunotherapy.py` provide a worked example [6, 32].

(c) `national-platform/ich_e6r3_adapt/01_preamble_principles_investigator.tex` ICH E6(R3) adaption and `02_sponsor_responsibilities.tex` already permits real-time consent updates as a recognized form of investigator-sponsor correspondence; the bill formalizes the patient-initiated form [39].

(d) The 21 CFR Part 50 adaption at `regulatory/` provides the Common Rule informed-consent language that HR 9503 builds on [40].

(e) Information-blocking penalties under the federal register disincentives rule provide the enforcement teeth for the bill’s five-minute logging requirement [28].

3.2. Definitions

(a) “*Procedural modification*” means a patient-initiated change to a pre-approved protocol step, executed by the qualified Physical AI system, that does not alter the primary endpoint or the safety profile beyond pre-defined tolerances.

(b) “*Real-time consent update*” means a HIPAA-compliant, FHIR-based consent revision logged in the patient’s electronic health record within five minutes of the modification. The FHIR infrastructure baseline is the MCP-PAI five-server topology with hash-chained audit trails at `national-platform/national_mcp/chunk2_methods_and_results_part1.tex` and `chunk3_results_part2.tex` [22].

(c) “*Qualified Physical AI executor*” means a robot or robotic system with a USL score above the modification-tier threshold (USL 8.0 or above for surgical modifications, 6.0 or above for therapeutic positioning modifications, 4.0 or above for rehabilitation modifications) [32].

(d) “*Modifiable steps matrix*” means a per-protocol matrix listing every step that the patient may modify under broad consent without requiring fresh IRB review. The matrix is delivered as part of the trial-enrollment package and is updated whenever the protocol itself is amended.

3.3. Patient Procedural Modification Rights

(a) Right to request, at any point during the trial, a modification to a pre-approved procedural step. The qualified Physical AI executor evaluates safety in real time and either executes or declines the modification with a cited reason logged in the patient’s record.

(b) Right to receive, in plain language, the safety evaluation result for any requested modification within five minutes. If the modification is declined, the patient may appeal to the IRB on call. IRB framework at `national-platform/21cfr50_adapt/02_irb_review_pediatric.tex` applies [40].

(c) Right to revise broad consent at any point with 24-hour notice, drawing on the OHRP broad-consent baseline [36].

(d) Right to specify, before each procedure, which steps may be modified by the patient and which steps require investigator review. The 10-stage patient journey at `patient-journey/` enumerates per-stage modification points across `stage_05_surgery.py` through `stage_10_closeout.py` [6].

(e) Right to request that a procedural step be performed by a Physical AI system rather than by a human doctor or nurse where doing so reduces the documented error rate. This right dovetails with HR 9504 and is enforceable through that bill’s error-rate publication requirement.

(f) Right to a per-cycle modifiable steps matrix delivered as part of the trial enrollment package, listing every step that the patient may modify under broad consent without requiring fresh IRB review.

3.4. Implementation

(a) The Secretary, acting through the Food and Drug Administration, the Office for Human Research Protections, and the Department of Health and Human Services, shall require federal IRB networks to publish a per-protocol modifiable-steps matrix within 18 months of enactment.

(b) `new-trial/site/02-legislation-patient-rights/physical_ai_patient_rights.tex` California baseline and `national-platform/new_trial_psl/02_ab2847_patient_rights.tex` are preserved by the preemption rule [21].

(c) The MCP-PAI infrastructure required by Subsection 3.2(b) shall be deployed at all CMS-billable Physical AI oncology trial sites within 24 months of enactment [22].

3.5. Reporting and Enforcement

(a) Annual federal report on modification rate by stage, by cancer, by robot category. Domain 2 of the federal Patient Control Dashboard (Decision Quality) anchors this report: consent comprehension, decisional-conflict scores, goal-trial concordance, AI-exclusion appeal success rate, and explanation and provenance coverage. The HTI-1 DSI transparency framework provides the technical baseline for each metric [23].

(b) `sponsor/final_paper/scripts/core_agents/audit_trail_manager.py` audit-trail manager provides the operational baseline for hash-chained provenance of modification requests [19].

(c) Civil penalties harmonize with 45 CFR 46 enforcement and the Cures Act [28].

3.6. Discussion and Limitation

The bill extends the prior 21st Century Cures Act broad-consent permissions into a positive procedural modification authority and extends the OHRP Broad Consent Recommendations 2017 into a real-time consent update framework [37, 41, 36]. The principal limitation is that the five-minute consent-update logging requirement assumes MCP-PAI infrastructure that is documented but not yet field-deployed at scale; the modifiable-steps matrix presumes IRB review at scale that has not been tested in production oncology trials [39].

3.7. Patient-Control Framings

For the individual patient, a 58-year-old female NSCLC patient matching PAT-2026-0042 in `patient-journey/stage_07_immunotherapy.py` requests at cycle 22 of 35 pembrolizumab that the Franka Emika cobot deliver the infusion 30 minutes earlier than the protocol default to accommodate work, with the qualified Physical AI executor evaluating safety in 4 minutes and approving the modification under the patient's broad-consent revision [6, 32].

For broad adoption, all Physical AI oncology trial sites must publish a modifiable-steps matrix per protocol within 18 months. The bill positions the United States as Number 1 in the world for patient-led procedural modification under physical-AI execution, replacing the prior all-or-nothing consent pattern under traditional IRB review with a granular per-step national standard [39, 6].

4. HR 9504 (2026) - Physical AI Clinical Error Reduction Act of 2026 (Adaption of HTI-1 DSI Final Rule 2023 and FDA AI Decision-Making Draft Guidance 2025)

HR 9504 is an *Adaption of the HTI-1 DSI Final Rule of 2023 and the FDA AI Regulatory Decision-Making Draft Guidance of January 2025*, drawing the chain of prior authority from the ONC algorithm-transparency rule through the FDA capability-neutrality draft [29, 23, 30]. The bill implements the sixth dimension of the operational definition of patient control introduced by this paper: AI-exclusion contestability with mandatory appeal rights, supplemented by the new error-rate publication requirement that converts HTI-1 DSI transparency into actionable patient-side substitution rights. The bill addresses the project brief's central aim of reducing human doctor and nurse error rates and the lack of physical and intellectual capabilities of human-only care, while preserving irreplaceable human roles in empathy, consent counseling, and ethical oversight.

4.1. Findings and Declarations

(a) Human doctors and nurses make errors at documented per-task rates published in peer-reviewed literature; the bill cites the prior published rates rather than inflating them. The literature covers surgical site infection, medication reconciliation, needle placement deviation, RT positioning, sepsis recognition, and adverse-event detection from electronic patient-reported outcomes [27].

(b) AI and robotics can reduce specific error categories where prospectively validated. The SHIELD-RT prospective randomized controlled trial reported a 45 percent reduction in acute-care events and a 48 percent reduction in associated costs in radiotherapy adverse-event prediction [27]. The ASyMS and eRAPID ePRO systems demonstrated patient-side adverse-event self-management with documented quality-of-life improvements [42, 43, 44].

(c) Computational pathology foundation models reach clinical-grade performance across pan-cancer detection. Virchow (Nature Medicine 2024) demonstrated this performance across both common and rare cancers, providing the technical baseline for AI second-opinion substitution [35].

(d) The Mazor 2025 randomized trial of AI notifications alone returned a null result on the trial-enrollment endpoint, indicating that AI alerts in isolation do not improve participation; the finding informs the bill's "no automated-only denial" guardrail [20].

(e) Bayesian urgent-care prediction (Gonzalez 2025) and the four author simulation accelerated patient-prediction paper at [new-trial/national-24-7-trial/paper/full-paper/](#) provide domain-specific evidence that AI patient prediction outperforms narrow supervised models on the specific tasks tested [45, 5].

(f) Humans retain irreplaceable roles in empathy, consent counseling, and ethical oversight. The bill does not eliminate human doctors or nurses; it shifts the patient's default for procedurally definable tasks to physical-AI execution where the patient prefers and the evidence supports.

(g) The FDA Real-Time Clinical Trials press announcement of April 28, 2026 establishes a regulatory opening that aligns sponsor-side real-time signal sharing with the patient-side error-rate substitution rights enacted by this bill [18].

4.2. Definitions

(a) “*Documented human error rate*” means the per-task error rate published in peer-reviewed literature for the specific task category (surgical site infection, medication reconciliation, needle placement deviation, RT positioning, sepsis recognition, adverse-event detection from ePRO).

(b) “*Physical AI substitute*” means a Physical AI system, robot, or agent that can perform the same task category with a documented error rate at or below the human baseline. The 53 core agents at `sponsor/final_paper/scripts/core_agents/` (in particular `safety_agent.py`, `clinical_accountability_agent.py`, and `audit_trail_manager.py`) provide the operational baseline [19].

(c) “*Patient-initiated AI second opinion*” means the patient’s right to request, before any consequential clinical decision, a parallel evaluation by a qualified AI system meeting HTI-1 DSI transparency requirements [23].

4.3. Error Rate Reduction Rights

(a) Right to a published per-site, per-task human error rate prior to consenting to any procedure.

(b) Right to substitute a Physical AI executor for any task where the documented Physical AI error rate is below the human baseline and the patient prefers AI execution.

(c) Right to a patient-initiated AI second opinion, generated by a qualified system, prior to any consequential clinical decision (surgery, chemotherapy regimen change, end-of-life care decision) [23].

(d) Right to mandatory error tracking across all sites, with public per-site reporting. The dashboard at `sponsor/final_paper/scripts/dashboard/` and the 168-hour cumulative metrics at `sponsor/final_paper/168_hours/instructions/core_i5_6200u_4gb/sponsor_168h_summary.json` provide the requisite operational reporting pattern [19].

(e) Right to refuse a human-only execution path for any task where a qualified Physical AI substitute exists, mirroring the FDA RTCT pilot’s “real-time signal sharing” model [18].

(f) Right to AI-exclusion contestability: any patient excluded from a trial, treatment, or procedural pathway by an automated decision may appeal under the HTI-1 DSI transparency framework [29, 23].

4.4. Comparison Diagram (Human Baseline vs Physical AI)

The following ASCII comparison diagram summarizes the per-task human-baseline and Physical AI error rates for the six task categories tracked under Subsection 4.5(a). Human baselines are sourced from the prior literature cited in `patients/paper/Deep-Research-1/chunk_1_foundations.md`

and the artificial intelligence baselines from `new-trial/national-24-7-trial/Background-A/sponsor/final_paper/scripts/output/sponsor_24h_summary.json`.

Task Category	Human Baseline (published)	Physical AI (qualified)
Surgical site infection	3 to 5 %	1 to 2 %
Medication reconciliation	10 to 15 %	2 to 4 %
Needle placement deviation	4 to 8 %	1 to 3 %
RT positioning	3 to 6 %	< 1 %
Sepsis recognition latency	3 to 6 hours	< 1 hour
AE detection from ePRO	30 to 40 %	< 10 %

The numbers above are illustrative ranges drawn from the cited literature and the author’s prior simulation results; site-specific rates are required to be published under Subsection 4.5(a).

4.5. Implementation

(a) The Secretary, acting through the Food and Drug Administration, the Agency for Healthcare Research and Quality, the Office of the National Coordinator for Health Information Technology, and the Centers for Medicare and Medicaid Services, shall require all CMS-billable sites to publish per-task human and AI error rates within 18 months of enactment.

(b) HTI-1 DSI transparency requirements apply to every AI second-opinion system deployed under Subsection 4.3(c) [23].

(c) `new-trial/national-24-7-trial/paper/full-paper/sections/discussion.tex` patient priority operational baseline informs the clinical trial implementation guidance [5].

4.6. Reporting and Enforcement

(a) Annual federal error-rate report by site, by task, and by AI substitute (where one is deployed).

(b) CMS reimbursement penalty for sites that fail to publish error rates within 18-months.

(c) Information-blocking penalty harmonized with the Cures Act disincentives rule [28].

(d) The federal Patient Control Dashboard’s Domain 3 (Control-in-Action) and Domain 4 (Fairness and Technical Quality) anchor the report. Domain 3 metrics: hospitalization rate, urgent-care or ED use, alert response time, false-alert burden, serious adverse events in home or robotics settings. Domain 4 metrics: declared context of use, external validation, subgroup performance, calibration monitoring, update-control disclosure, revalidation [27, 45, 17, 33, 30, 35].

4.7. Implementation Guardrails

The bill is bound by three guardrails that anchor the bills paper-wide and that are restated in HR 9506 for the cross-bill leadership authority.

- (i) No patient denied trial consideration by automated output alone.
- (ii) Capability-neutral statutory rights that follow any AI architecture (LLM, foundation model, agentic system, robotics platform), drawing on the FDA AI Regulatory Decision-Making Draft [30].
- (iii) Mandatory annual public reporting, auditability, and sunset review by Congress under FDORA Section 3209 authorities [26].

4.8. Discussion and Limitation

The bill extends ONC HTI-1 DSI transparency into mandatory error-rate publication and patient-initiated AI second-opinion rights. The ePRO and remote symptom monitoring evidence (Rocque 2025, ASyMS, eRAPID) supports the bill without overclaiming. The Mazor 2025 null result on AI notifications alone is critical and is incorporated into the "no automated-only denial" guardrail [29, 27, 42, 43, 44, 20]. The bill is the most contentious of the seven because it touches workforce questions beyond the narrow project brief; the Huang 2025 null result indicates that machine learning does not always exceed Cox regression on real-world structured survival data; SHIELD-RT is convincing for radiotherapy but does not generalize without prospective validation in each domain. The bill respects human doctor and nurse roles in empathy, consent counseling, and ethical oversight, and constrains the AI-substitution path to procedurally definable tasks with prospectively validated error reductions [5, 6].

4.9. Patient-Control Framings

For the individual patient, a 58-year-old female NSCLC patient matching PAT-2026-0042 reviews the published per-site error rates at her chosen site, requests an AI second opinion before her cycle-1 pembrolizumab dosing decision, and refuses a human-only RT positioning step in favor of the RTPOS-01 robot at hour-12 of `new-trial/national-24-7-trial/hour-12/robot_logs.md` [5, 6].

For broad adoption, all United States CMS-billable oncology trial sites must publish per-task error rates within 18 months. The bill positions the United States as Number 1 in the world for evidence-based patient-led error reduction. The bill explicitly respects the human doctor and nurse roles that remain irreplaceable (empathy, consent counseling, ethical oversight) and constrains the AI-substitution path to definable tasks with prospectively validated error reductions [5, 6].

5. HR 9505 (2026) - Real-Time Patient-Sponsor Direct Communication Act of 2026 (Revision of FDA Real-Time Clinical Trials Press Announcement April 2026 and FDA Decentralized Clinical Trials Guidance 2024)

HR 9505 is a *Revision of the FDA Real-Time Clinical Trials Press Announcement of April 28, 2026 and the FDA Decentralized Clinical Trials Final Guidance of 2024*, drawing the chain of prior authority from the FDA’s own validation of the Paradigm-Health-style real-time signal-sharing conduit through the FDA decentralized- elements final guidance [18, 2, 46]. The bill extends a sponsor-initiated proof of concept (TRAVERSE Phase 2 in mantle cell lymphoma at MD Anderson and Penn; STREAM-SCLC Phase 1b at Mass General) into a patient-initiated direct sponsor communication right, while remaining respectful of the FDA’s leadership posture as the agency the bill builds on rather than against. The bill implements the fifth dimension of the operational definition of patient control: toxicity warnings and adverse-event self-management through ePRO, ASyMS, and eRAPID infrastructure [27, 42, 43, 44].

5.1. Findings and Declarations

(a) FDA Commissioner Marty Makary, in the April 28, 2026 press announcement, stated that "for 60 years, we’ve been conducting clinical trials in the same way" and framed real-time clinical trials as a transformative shift in how trials are run. FDA Chief AI Officer Jeremy Walsh emphasized the transformative potential of real-time trials in the same announcement [18].

(b) Two announced proof-of-concept trials anchor the FDA’s validation: AstraZeneca’s TRAVERSE Phase 2 trial in mantle cell lymphoma at MD Anderson and the University of Pennsylvania, and Amgen’s STREAM-SCLC Phase 1b trial at Massachusetts General Hospital [18].

(c) Paradigm Health is the validated technical conduit for real-time signal sharing in both proof-of-concept trials [18].

(d) The author’s prior 168-hour autonomous sponsor simulation at repository file location `sponsor/final_paper/168_hours/` demonstrated that hourly commits from a sponsor-side server (24 repository commits per day, 168 commits across seven days) sustain a reduced size trial-data flow on minimal commodity hardware (Core i5-6200U with 4 GB RAM, see `sponsor/final_paper/168_hours/instructions/core_i5_6200u_4gb/README.md`) [19].

(e) The ePRO and remote symptom monitoring evidence (Rocque 2025; ASyMS; eRAPID) supports the bill without overclaiming [27, 42, 43, 44].

(f) The FDA pilot is sponsor-initiated; this bill extends the same real-time channel to patient-initiated communication while preserving the FDA’s regulatory leadership.

5.2. Definitions

(a) “*Patient-sponsor direct channel*” means a HIPAA-compliant, FHIR-based, Paradigm-Health-style aggregator endpoint that the patient may use to send the patient’s own structured data, modification requests, and adverse-event reports directly to the trial sponsor’s safety

pharmacovigilance system. The operational baseline is the FastAPI sponsor control server at `sponsor/final_paper/scripts/sponsor_server/` and the safety-pharmacovigilance section of the prior sponsor paper at `sponsor/paper/sections/safety_pharmacovigilance.tex` [19].

(b) “*Real-time*” means data from the patient’s qualified Physical AI executor, ePRO endpoint, or wearable reaches the sponsor within one hour of generation. The hourly commit cadence documented in the four-simulation paper at `new-trial/national-24-7-trial/paper/sections/results.tex` (Simulation 1: 24 commits per day; 84 commits across 84 hours) and the 168-hour extension at `sponsor/final_paper/168_hours/` (Simulation 4: 168 GitHub commits across seven days) provide the operational baseline [5, 19].

(c) “*Sponsor-side patient interface*” means the cancer patient facing endpoint that mirrors `sponsor/final_paper/scripts/sponsor_server/` but is accessible to the patient under HIPAA-compliant authentication.

5.3. Patient-Sponsor Direct Communication Rights

(a) Right to send structured adverse-event reports (Common Toxicity Criteria graded) directly to the sponsor’s pharmacovigilance system within one hour, drawing on the ePRO infrastructure documented in Rocque 2025 and the ASyMS / eRAPID systems [27, 42, 43, 44].

(b) Right to receive sponsor-side acknowledgment of every patient submission within one hour, with the acknowledgment carrying a hash-chained provenance record per the MCP-PAI standard documented at `national-platform/national_mcp/` [22].

(c) Right to receive a per-cycle digital twin update reflecting the patient’s own data submissions. The technical baseline is the digital-twin initialization Python module at repository file location `patient-journey/stage_03_digital_twin.py` and the patient-modeling framework at `digital-twins/patient-modeling/` [6].

(d) Right to escalate any sponsor-side delay beyond one hour to the FDA RTCT pilot oversight team, mirroring the escalation pattern at `sponsor/final_paper/scripts/coordination/` [19].

(e) Right to receive a real-time read-back of how the patient’s data is being used in the sponsor’s safety analyses, with the privacy envelope governed by 45 CFR 46 and the data-protection framework in HR 9507 [1].

(f) Right to opt out of any individual sponsor-data exchange while remaining enrolled in the trial. The patient retains discretion over which fields the sponsor sees and may withdraw any individual field without withdrawing from the trial.

5.4. Implementation

(a) The Secretary of Health and Human Services, acting through the Food and Drug Administration, the Office of the National Coordinator for Health Information Technology, and the Department of Health and Human Services, shall require all FDA RTCT pilot trials to expose a patient-sponsor direct channel within 24 months of enactment.

(b) Implementation aligns with the FDA’s own RTCT pilot program and the FDA Oncology Center of Excellence Advancing Oncology Decentralized Trials directory [3].

(c) The NIH funding opportunity for digital-health-technology-derived endpoints in cancer trials (PAR-25-170) supplies the research-funding bridge to expand the patient-sponsor direct channel beyond the initial pilot [47].

5.5. Reporting and Enforcement

(a) Annual federal report on patient-sponsor channel uptake and median sponsor acknowledgment latency. The federal Patient Control Dashboard’s Domain 3 (Control-in-Action) subset anchors this report: alert response time, false-alert burden, median sponsor acknowledgment latency, and serious adverse events detected via patient-initiated channel versus sponsor-initiated channel [27].

(b) Civil penalties for sponsors that fail to acknowledge within one hour.

(c) `sponsor/final_paper/scripts/core_agents/audit_trail_manager.py` audit-trail provenance model provides the technical baseline for hash-chained provenance records [19].

5.6. Discussion and Limitation

The ePRO and remote symptom monitoring evidence supports the proposed new bill without overclaiming. The author’s prior simulation evidence (consisting of the four LLM simulations in `new-trial/national-24-7-trial/paper/full-paper/`, the single-patient journey at `patient-journey/`, and the 24-hour and 168-hour sponsor simulations in `sponsor/final_paper/`) supplies the operational substrate that the bill regulates. The FDA’s "demonstrated proof-of-concept" framing (TRAVERSE, STREAM- SCLC, Paradigm Health) is the regulatory opening that HR 9505 builds on [5, 6, 19, 18, 46]. The principal limitation is that Paradigm-Health-style aggregator infrastructure has not yet been demonstrated to generalize beyond TRAVERSE and STREAM-SCLC; the FDA RTCT pilot has not released aggregate metrics; and the 24-month deadline is aggressive [18].

5.7. Patient-Control Framings

For the individual patient, a 58-year-old female NSCLC patient matching PAT-2026-0042 sends a Grade 2 fatigue adverse-event report from her home wearable on day 14 of cycle 22 of her 35 pembrolizumab cycles. The sponsor’s safety pharmacovigilance system acknowledges the report within 12 minutes and updates her per-cycle digital twin within the same hour. The patient reviews the read-back showing how her adverse-event data informed the cumulative safety analysis at `sponsor/final_paper/scripts/output/sponsor_24h_summary.json` [6, 19].

For broad adoption, all FDA RTCT pilot trials must expose a patient-sponsor direct channel within 24 months. The bill positions the United States as Number 1 in the world for patient-initiated real-time trial communication, building respectfully on the FDA’s own April 28, 2026 RTCT framework and the 168-hour autonomous sponsor simulation in `sponsor/final_paper/168_hours/` [19, 5, 18].

6. HR 9506 (2026) - American Physical AI Oncology Leadership Act of 2026 (New Federal Authorizing Statute Adapting FDORA Section 3209 and 21st Century Cures Act 2016)

HR 9506 is a *New Federal Authorizing Statute Adapting FDORA Section 3209 (PL 117-328) and the 21st Century Cures Act of 2016 (PL 114-255)*, drawing the chain of prior authority from the in silico authorization in FDORA through the interoperability and information-blocking framework of the Cures Act [26, 37]. The bill is the project brief's "United States Number 1" statute and absorbs the cross-cutting paper-level material from the prior template's Implementation Metrics, Discussion, Limitations, and Conclusions sections. The respectful FDA framing throughout this section is intentional: the bill is an authorization, not a critique.

The current legislative gap closed by HR 9506 is the absence of a standalone federal authorizing statute for sustained American leadership in medical AI and robotics oncology. FDORA Section 3209 already authorizes in silico and computer-model alternatives to animal testing, mandates diversity action plans, and mandates the FDA decentralized-trial guidance [26, 2]. HR 9506 extends that authorization from animal-testing alternatives to physical-AI executor selection and codifies the American Leadership Index that anchors the cross-bill implementation timeline.

6.1. Cross-Bill Implementation Timeline

The American Leadership Index publication cycle is anchored to the following cross-bill timeline. Use the diagram below as the canonical schedule for the seven bills (HR 9501 through HR 9507).

Month	Bill	Federal Deadline
12	HR 9501	Federal API for self-selection
12	HR 9502	Per-robot USL transparency reports
12	HR 9506	First American Leadership Index publication
12	HR 9507	CMS-billable EHR self-custody endpoint
18	HR 9503	Per-protocol modifiable steps matrix
18	HR 9504	Per-task human and AI error rates
24	HR 9501	Full state-level self-selection adoption
24	HR 9505	Patient-sponsor direct channel coverage
24	HR 9506	FDA RTCT pilot expansion (target 25)
24	HR 9507	State-level data self-custody harmonization
36	HR 9506	All 50 states with at least one Physical AI oncology trial site authorized
84	HR 9506	Seven-year sustained federal investment period 2026 through 2032 ends

6.2. Findings and Declarations

(a) The United States is the global leader in clinical-trial infrastructure today and must remain Number 1 in the world for patient benefit during the medical AI and robotics revolution [22, 48].

(b) The FDA Real-Time Clinical Trials announcement of April 28, 2026 and the FDA Oncology Center of Excellence Advancing Oncology Decentralized Trials directory provide evidence of FDA leadership posture that this bill formalizes into statute [18, 3].

(c) FDORA Section 3209 already authorizes in silico alternatives; HR 9506 extends that authorization from animal-testing alternatives to physical-AI executor selection and to the federated-learning infrastructure across CMS-billable sites [26].

(d) The 21st Century Cures Act (PL 114-255) supplies the interoperability and information-blocking framework that the American Leadership Index reports against [37].

(e) The author’s prior repository deposits document the technical substrate at site, sponsor, patient-journey, patient-instructions, USL, and ICH-E6(R3) levels and supply the verifiable basis for the American Leadership Index reporting metrics [21, 6, 19, 31, 32, 39].

(f) Foundation-model evidence (Virchow), trial-matching evidence (TrialGPT, PRISM, Trial-MatchAI), and prediction evidence (SCORPIO, PROGPATh, the four-simulation paper at `new-trial/national-24-7-trial/paper/full-paper/`) anchor the bill’s authorization for investment in foundation models across pathology, prediction, and trial matching [35, 9, 10, 11, 5].

6.3. Definitions

(a) “*Medical AI and robotics revolution*” means the period from approximately 2024 through 2030 during which Physical AI systems and large-context AI agents (for example Claude Code Opus 4.7 with 1M token context, see [49]) are entering routine clinical-trial deployment. The four-simulation evidence base at `new-trial/national-24-7-trial/paper/full-paper/sections/results.tex` provides the operational reference [5].

(b) “*American leadership*” means a documented United States lead in: (i) the number of authorized Physical AI oncology trial sites; (ii) the number of patient-self-selected trial bookings per year; (iii) the cumulative documented error-rate reduction across CMS- billable sites; (iv) the number of patients whose data is governed under HR 9507 self-custody; and (v) the number of FDA RTCT pilot trials with patient-sponsor direct channels under HR 9505.

(c) “*Federal investment*” means dedicated appropriations to the Food and Drug Administration, the National Institutes of Health (PAR-25-170 and successor programs, see [47]), the Agency for Healthcare Research and Quality, and the Office of the National Coordinator for Health Information Technology for the seven-year period 2026 through 2032.

(d) “*American Leadership Index*” means the public report published annually by the FDA Oncology Center of Excellence with the five metrics defined in 6.3(b) above.

6.4. American Leadership Authorizations

(a) Authorization for the FDA RTCT pilot program to expand from the initial two trials (TRAVERSE, STREAM-SCLC) to at least 50 trials by 2028 [18].

(b) Authorization for the FDA Oncology Center of Excellence to publish per-state dashboards of Physical AI oncology clinical trial site authorization counts, harmonized with the prior trial site documentation framework at `new-trial/site/05-state-regulations/` and at `national-platform/new_trial_psl/05_title22_regulations.tex` [21, 22].

(c) Authorization for federated-learning infrastructure across CMS-billable sites, drawing on the framework at `national-platform/federated_learning/` (chunks 1 through 4) [22].

(d) Authorization for federated investment in foundation models for pathology (Virchow), prediction (SCORPIO, PROGPATH), and trial matching (TrialGPT, PRISM, TrialMatchAI), with annual per-model reporting on patient-benefit metrics [35, 9, 10, 11].

(e) Authorization for a public American Leadership Index published annually with the five metrics defined in 6.3(b).

6.5. New Physical-AI Reversal Guardrail

A new physical-AI guardrail introduced by this paper is anchored under HR 9506’s annual American Leadership Index reporting authority: physical-AI executor selections and procedural modifications must always permit a patient-initiated reversal to human-only execution within the same calendar day. The operational baseline is `patient-journey/master_journey.py` and the manager at `sponsor/final_paper/scripts/core_agents/audit_trail_manager.py` [6, 19].

6.6. Implementation

(a) The Secretary of Health and Human Services, acting through the Food and Drug Administration, the National Institutes of Health, the Agency for Healthcare Research and Quality, and the Office of the National Coordinator for Health Information Technology, shall publish the first American Leadership Index within 12 months of enactment.

(b) The FDA RTCT pilot shall expand to at least 25 trials within 24 months of enactment, with the target of 50 trials by 2028 [18].

(c) `national-platform/new_template/sections/implementation_strategy.tex` implementation pattern informs the implementation guidance [22].

6.7. Reporting and Enforcement

(a) Annual American Leadership Index report.

(b) Congressional review under FDORA authorities; explicit alignment with the FDORA Section 3209 in silico authorization [26].

6.8. Future Work Tracks

HR 9506 authorizes federal investment that supports three future tracks for delivering all seven bills’ obligations.

Track A (Single big model performing all bills’ obligations). A future generation of Claude Code Opus 4.7 (or equivalent foundation-model agent) holds all seven bill texts plus the AVAILABLE DIRECTORIES content in a single 1M-token context and emits implementation deliverables (machine-readable eligibility reports, USL transparency reports, modifiable-steps matrices, error-rate publications, real-time aggregator endpoints, the American Leadership Index, and self-custody endpoints) in a single inference loop. This track is the natural extension of the four-simulation paper at `new-trial/national-24-7-trial/paper/full-paper/` [5, 49].

Track B (Distributed agents per bill, coordinated by a smaller orchestrator). Each of the seven bills has its own dedicated agent (self-selection agent for HR 9501, robot-choice agent for HR 9502, modification agent for HR 9503, error-rate agent for HR 9504, sponsor-channel agent for HR 9505, leadership-index agent for HR 9506, self-custody agent for HR 9507), all coordinated by a smaller orchestrator that runs on hardware as constrained as the Core i5-6200U with 4 GB RAM at `sponsor/final_paper/168_hours/instructions/core_i5_6200u_4gb/README.md` [19].

Track C (Patient-side native deployment). A future generation deploys a single integrated patient-side agent that operates HR 9501 through HR 9507 from the patient’s own device, using patient-controlled credentials and HIPAA-compliant authentication to interact with provider, payer, and sponsor endpoints. This track is the natural extension of `patients/patient_robot_instructions_fixed.tex` pages 1 through 10 plus the digital-twin infrastructure at `patient-journey/stage_03_digital_twin.py` [31, 6].

6.9. Future Work Deliverables

HR 9506 authorizes federal investment supporting three concrete future deliverables: (i) a TRIPOD+AI / CREMLS-compliant retrospective validation of HR 9504’s error-rate baseline on a real-patient cohort, drawing on the SHIELD-RT design pattern; (ii) a public benchmark of HR 9501’s machine-readable eligibility report performance across the TrialGPT, PRISM, and TrialMatchAI matching systems on a shared dataset; and (iii) an FDA-aligned RTCT pilot submission that uses Simulation 4 (`sponsor/final_paper/168_hours/`) as the sponsor-side input and HR 9505’s patient-sponsor direct channel as the patient-side input [18, 5, 19, 6].

6.10. Persistent Themes Across the Seven Bills

The persistent themes anchored under HR 9506’s American Leadership Index are: (1) Adaption / Revision framing, with every bill explicitly stating the prior United States legislative document it builds from; (2) respectful FDA framing, with every bill positioning the FDA and other governing bodies as enabling rather than gating while remaining opportunistic under the patient’s control based primarily on the patient’s own interests; (3) individual patient plus broad-adoption framing,

with every bill naming per-patient examples (PAT-2026-0042 from `patient-journey/`) AND broad-adoption deadlines (12, 18, 24 months across CMS-billable sites and 50 states); and (4) United States Number 1, with every bill closing with the leadership framing required to keep the United States Number 1 in the world for patient benefit during the medical AI and robotics revolution.

6.11. Implications for Cancer Patient Control

Large language model-scale repository context with one million tokens capacity demonstrated in `new-trial/national-24-7-trial/paper/full-paper/`, the four LLM simulations summarized at [5], the 168-hour sponsor extension at `sponsor/final_paper/168_hours/`, the 10-stage single-patient journey at `patient-journey/`, and the 10-robot-category patient instructions at `patients/patient_robot_instructions_fixed.tex` together show that the technical substrate is ready. The seven bills supply the missing legal substrate. Together they enable the cancer patient to take control of the patient's disease in ways that prior error-prone and physically and intellectually limited human-only care could not support.

6.12. Discussion and Limitation

The bill is the most ambitious of the seven; it assumes seven-year sustained federal investment which is subject to appropriations cycles. The American Leadership Index relies on year-over-year metric stability across 50 states. Across the seven bills, the draft language is by an independent author and has not been reviewed by Congress, the FDA, or any State legislature. The bills are not endorsed by Anthropic, OpenAI, Google, or any technology vendor. The patient-control framework remains an operational hypothesis; outcome studies are required to confirm that the bills, taken together, increase participation and decrease error rates without unintended consequences. The bills do not address pediatric or vulnerable-population protections beyond the existing 21 CFR 50 Subpart D at `national-platform/21cfr50_adapt/02_irb_review_pediatric.tex` [26, 40].

6.13. Forward Path

The next Claude Code Opus 4.7 1M Max generation pass will refine this paper with additional ASCII diagrams, additional individual patient and robot examples drawn from the AVAILABLE DIRECTORIES, and additional cross-bill cross-references; the author will then perform additional senior-author white-space and table-column-width formatting passes to remove any orphans, widows, or large empty page regions before publication. References anchor the forward path: [7], [18], [5], [19], [22], and [31].

6.14. Patient-Control Framings

For the individual patient, a cancer patient anywhere in the United States can verify, in the published American Leadership Index, that the patient's State has at least one authorized Physical AI oncology trial site, that the per-task error rates have improved year-over-year, and that the FDA

RTCT pilot covers the patient’s cancer type. PAT-2026-0042 in `patient-journey/` illustrates the per-patient outcome trajectory across the 10-stage journey [6].

For broad adoption, all 50 states must have at least one authorized Physical AI oncology trial site within 36 months of enactment; the United States retains Number 1 status in the world for patient benefit during the medical AI and robotics revolution by virtue of the per-state, per-cancer, per-site dashboard publication; federal appropriations sustain the FDA, NIH, AHRQ, and ONC infrastructure for seven years [22, 5, 19].

7. HR 9507 (2026) - Cancer Patient Health Data Self-Custody and Trial-Selection Act of 2026 (Revision of HHS HIPAA Right-to-Access 2025 and ONC Cures Act Final Rule)

HR 9507 is a *Revision of the HHS HIPAA Right-to-Access (2025) framework and the ONC Cures Act Final Rule*, drawing the chain of prior authority from the HIPAA right-to-access right through the information-blocking and certification provisions of the Cures Act [14, 41, 28]. The bill implements the fourth dimension of the operational definition of patient control: data and biospecimen governance under HIPAA, broad consent, and information-blocking penalties.

The current legal stack supplies the foundation. The 21st Century Cures Act (PL 114-255) established interoperability, information-blocking, real-world evidence, and the non-device clinical decision support software carve-out. HHS HIPAA Right-to-Access (2025), the ASTP/ONC Patient Portals Brief 2024, and the ASTP/ONC Hospital APIs Brief 2024 establish the access framework. Information-blocking penalties exist under the federal register disincentives rule but patient-initiated self-custody of eligibility data is not yet guaranteed. HR 9507 closes that gap [37, 41, 14, 15, 16, 28].

7.1. Findings and Declarations

(a) Patient access to health information is a recognized HIPAA right; the right does not by itself create a patient-controlled self-custody endpoint suitable for direct grant of trial-sponsor or AI-matching access [14].

(b) The ASTP/ONC Patient Portals and Hospital APIs briefs document that adoption of patient-controlled access is uneven across health-system EHR vendors [15, 16].

(c) Foundation-model performance in computational pathology (Virchow, Nature Medicine 2024) supports HR 9507's same-day pathology release requirement: pathology slides are the dominant input for trial-eligibility gating in solid tumors, and timely delivery of high-quality slides materially improves the trial-matching pipeline [35].

(d) The prior California SB 892 Clinical Data Protection and Transparency Act (documented at `new-trial/site/03-legislation-data-transparency/physical_ai_data_transparency.tex` and at repository location `national-platform/new_trial_psl/03_sb892_data_protection.tex`) provides the State-level baseline that HR 9507 revises into federal coverage [21, 22].

(e) The information-blocking enforcement record is mixed; HR 9507 incorporates the federal register disincentives rule [28].

7.2. Definitions

(a) *"Patient health data self-custody"* means the patient's right to hold the patient's own complete oncology medical record, in machine-readable FHIR format, on a patient-controlled device or cloud endpoint, with full HIPAA portability across providers, payers, and trial sponsors.

(b) *"Trial-eligibility data"* means the subset of the patient's medical record that determines eligibility for currently recruiting oncology trials, including pathology slides and reports, genomic

sequencing reports, prior treatment history, ECOG / PD-L1 / TMB scores, and current medications. The Virchow pathology benchmark supports the inclusion of slide-level data, and the digital-twin module at `patient-journey/stage_03_digital_twin.py` provides the technical baseline [35, 6].

(c) “*Same-day release*” means pathology and genomic results released to the patient’s self-custody endpoint within the same calendar day the result is finalized, drawing on Domain 1 of the four-domain dashboard at `patients/paper/Deep-Research-3/part3_metrics_guardrails.md`.

7.3. Patient Health Data Self-Custody Rights

(a) Right to receive the complete medical record in machine-readable FHIR format on demand, at zero cost to the patient.

(b) Right to require same-day release of pathology and genomic results to the patient’s self-custody endpoint, drawing on the ASTP/ONC hospital APIs framework [16].

(c) Right to grant time-limited, scope-limited access to specific trial sponsors or AI matching systems (TrialGPT, PRISM, TrialMatchAI), with revocation within one hour [9, 10, 11].

(d) Right to a portable oncology clinical trial-history document record that follows the cancer patient across multiple different providers, payers, and States. The data-governance baseline at `national-platform/ich_e6r3_adapt/03_data_governance.tex` applies [39].

(e) Right to information-blocking penalties under the federal register disincentives rule where any provider or payer denies the patient’s self-custody request [28].

(f) Right to a permanent, exportable copy of the complete oncology medical record at end of trial enrollment, with no provider retention requirement that overrides self-custody rights.

7.4. Implementation

(a) The Secretary of Health and Human Services, acting through the Food and Drug Administration, the Office of the National Coordinator for Health Information Technology, the Office for Civil Rights, and the Centers for Medicare and Medicaid Services, shall require all CMS-billable EHR vendors to publish a self- custody export endpoint within 12 months of enactment.

(b) State-level harmonization shall be complete within 24 months of enactment.

(c) The preemption rule preserves stronger State-level data- protection statutes, including the California baseline [21].

7.5. Reporting and Enforcement

(a) Annual federal report on per-state, per-payer self-custody uptake.

(b) Civil penalties under HIPAA and the information-blocking disincentives rule [28].

(c) CMS reimbursement penalty for sites that fail to expose the self-custody endpoint by the 12-month deadline.

(d) The federal Patient Control Dashboard’s Domain 1 (Access) subset anchors the report: same-day pathology release rate, same-day genomic release rate, end-to-end FHIR export latency,

patient- initiated trial sponsor access grants per quarter, and patient- initiated revocations within one hour [14, 16].

7.6. Discussion and Limitation

The bill extends the prior 21st Century Cures Act information-blocking framework into a positive self-custody right and extends the HHS HIPAA Right-to-Access 2025 framework with a same-day release requirement for pathology and genomic results. The Virchow foundation-model evidence supports the same-day pathology release because pathology data are dispositive for solid-tumor trial eligibility [37, 41, 35, 14]. The principal limitation is that the 12-month CMS-billable EHR self-custody endpoint deadline assumes vendor support at zero cost to the patient; the Cures Act information-blocking enforcement record is mixed [28].

7.7. Patient-Control Framings

For the individual patient, a 58-year-old female NSCLC patient matching PAT-2026-0042 receives her PD-L1 65 percent and TMB 14 mutations per megabase scores released to her self-custody endpoint on the same calendar day. She grants 30-day time-limited access to TrialGPT and TrialMatchAI for trial discovery (HR 9501), and revokes that access within one hour after self-booking her preferred trial [9, 11, 6].

For broad adoption, all CMS-billable EHRs must expose a self- custody export endpoint within 12 months; all United States cancer patients gain portable trial-eligibility data within 24 months. The bill positions the United States as Number 1 in the world for patient health data self-custody, replacing the prior access-only HIPAA framework with a positive self-custody right [6, 5].

References

- [1] U.S. Department of Health and Human Services. *45 CFR 46: Protection of Human Subjects (Common Rule)*. 2026. URL: <https://tinyurl.com/3cjvayc3>.
- [2] U.S. Food and Drug Administration. *Conducting Clinical Trials with Decentralized Elements*. Sept. 2024. URL: <https://tinyurl.com/3hx55rv6>.
- [3] U.S. Food and Drug Administration, Oncology Center of Excellence. *Advancing Oncology Decentralized Trials*. June 2024. URL: <https://tinyurl.com/y8ufd6np>.
- [4] United States House of Representatives, Office of the Law Revision Counsel. *42 U.S.C. 300gg-8: Coverage for Individuals Participating in Approved Clinical Trials*. 2026. URL: [https://uscode.house.gov/view.xhtml?req=\(title:42%20section:300gg-8%20edition:prelim\)](https://uscode.house.gov/view.xhtml?req=(title:42%20section:300gg-8%20edition:prelim)).
- [5] Kevin Kawchak. *Accelerated Patient Prediction in Physical AI Oncology Clinical Trials: Four Comprehensive LLM Simulations*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/new-trial/national-24-7-trial/paper>. May 2026. DOI: 10.5281/zenodo.19994945. URL: <https://doi.org/10.5281/zenodo.19994945>.
- [6] Kevin Kawchak. *A Cancer Patient's Journey Through a Regulated Physical AI Oncology Trial*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/patient-journey>. Mar. 2026. DOI: 10.5281/zenodo.19119939. URL: <https://doi.org/10.5281/zenodo.19119939>.
- [7] Kevin Kawchak. *Patient Priority and Proposed U.S. Bills for Physical AI Oncology Clinical Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/patients/paper>. May 2026. DOI: 10.5281/zenodo.20045457. URL: <https://doi.org/10.5281/zenodo.20045457>.
- [8] United States Congress. *CLINICAL TREATMENT Act / Consolidated Appropriations Act 2021, PL 116-260 (2020)*. 2020. URL: <https://www.congress.gov/bill/116th-congress/house-bill/133>.
- [9] Qiao Jin et al. "Matching Patients to Clinical Trials with Large Language Models". In: *Nature Communications* 15 (2024), p. 9074. DOI: 10.1038/s41467-024-53081-z. URL: <https://doi.org/10.1038/s41467-024-53081-z>.
- [10] Shashi Gupta et al. "PRISM: Patient Records Interpretation for Semantic Clinical Trial Matching System Using Large Language Models". In: *npj Digital Medicine* 7 (2024), p. 305. DOI: 10.1038/s41746-024-01274-7. URL: <https://doi.org/10.1038/s41746-024-01274-7>.

- [11] M. Abdallah et al. “TrialMatchAI: An End-to-End AI-Powered Clinical Trial Recommendation System to Streamline Patient-to-Trial Matching”. In: *Nature Communications* (2026). DOI: 10.1038/s41467-026-70509-w. URL: <https://doi.org/10.1038/s41467-026-70509-w>.
- [12] United States Congress. *Patient Protection and Affordable Care Act, PL 111-148 (2010), Section 2709*. 2010. URL: <https://www.congress.gov/bill/111th-congress/house-bill/3590>.
- [13] United States Congress. *Food and Drug Administration Amendments Act (FDAAA), PL 110-85 (2007), Section 801*. 2007. URL: <https://www.congress.gov/bill/110th-congress/house-bill/3580>.
- [14] U.S. Department of Health and Human Services, Office for Civil Rights. *Get It. Check It. Use It. You Have the Right to Access Your Health Information*. May 2025. URL: <https://www.hhs.gov/hipaa/for-individuals/right-to-access/index.html>.
- [15] Assistant Secretary for Technology Policy and Office of the National Coordinator for Health Information Technology. *Individuals’ Access and Use of Patient Portals and Smartphone Health Apps, 2024*. July 2025. URL: <http://healthit.gov/data/data-briefs/individuals-access-and-use-patient-portals-and-smartphone-health-apps-2024/>.
- [16] Assistant Secretary for Technology Policy and Office of the National Coordinator for Health Information Technology. *Hospital Use of APIs to Enable Data Sharing between EHRs and Third-Party Technology*. 2026. URL: <https://tinyurl.com/2s5sdhmn>.
- [17] Roxana S. Dronca, Jeremy C. Jones, et al. “Cancer CARE (Connected Access and Remote Expertise) Beyond Walls: Safety, Feasibility, and Patient Experience of Home-Based Chemotherapy”. In: *NEJM Catalyst Innovations in Care Delivery* 7.4 (2026). DOI: 10.1056/CAT.25.0168. URL: <https://doi.org/10.1056/CAT.25.0168>.
- [18] U.S. Food and Drug Administration. *FDA Announces Major Steps to Implement Real-Time Clinical Trials*. Apr. 2026. URL: <https://www.fda.gov/news-events/press-announcements/fda-announces-major-steps-implement-real-time-clinical-trials>.
- [19] Kevin Kawchak. *Fully Automated Sponsor: Physical AI Oncology Clinical Trial Platform*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/sponsor>. Apr. 2026. DOI: 10.5281/zenodo.19396256. URL: <https://doi.org/10.5281/zenodo.19396256>.
- [20] Tali Mazor, Karim S. Farhat, Pavel Trukhanov, et al. “Clinical Trial Notifications Triggered by Artificial Intelligence Detected Cancer Progression: A Randomized Trial”. In: *JAMA Network Open* 8.4 (2025), e252013. DOI: 10.1001/jamanetworkopen.2025.2013. URL: <https://doi.org/10.1001/jamanetworkopen.2025.2013>.

- [21] Kevin Kawchak. *Physical AI Oncology Clinical Trial Site Complete Documentation Package*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/new-trial/site>. Mar. 2026. DOI: 10.5281/zenodo.19176370. URL: <https://doi.org/10.5281/zenodo.19176370>.
- [22] Kevin Kawchak. *National Platform for Physical AI Oncology Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/national-platform>. Mar. 2026. DOI: 10.5281/zenodo.19244918. URL: <https://doi.org/10.5281/zenodo.19244918>.
- [23] Assistant Secretary for Technology Policy and Office of the National Coordinator for Health Information Technology. *Decision Support Interventions Fact Sheet: Health Data, Technology, and Interoperability (HTI-1) Final Rule*. Dec. 2023. URL: https://www.healthit.gov/wp-content/uploads/2023/12/HTI-1_DSI_fact-sheet_508.pdf.
- [24] Centers for Medicare and Medicaid Services. *National Coverage Determination 310.1: Routine Costs in Clinical Trials*. 2026. URL: <https://www.cms.gov/medicare-coverage-database/view/ncd.aspx>.
- [25] California State Legislature. *California SB 37 (2001) - Cancer Clinical Trial Routine Costs Coverage*. 2001. URL: https://leginfo.ca.gov/faces/billNavClient.xhtml?bill_id=200120020SB37.
- [26] United States Congress. *Food and Drug Omnibus Reform Act (FDORA) and FDA Modernization Act 2.0, PL 117-328 (2022), Section 3209*. 2022. URL: <https://www.congress.gov/bill/117th-congress/house-bill/2617>.
- [27] Gabrielle B. Rocque, Jeffrey A. Franks, Luqin Deng, et al. “Remote Symptom Monitoring With Electronic Patient-Reported Outcomes in Clinical Cancer Populations”. In: *JAMA Network Open* 8.5 (2025), e259852. DOI: 10.1001/jamanetworkopen.2025.9852. URL: <https://doi.org/10.1001/jamanetworkopen.2025.9852>.
- [28] U.S. Department of Health and Human Services. *21st Century Cures Act: Establishment of Disincentives for Health Care Providers That Have Committed Information Blocking*. July 2024. URL: <https://www.federalregister.gov/documents/2024/07/01/2024-13793/21st-century-cures-act-establishment-of-disincentives-for-health-care-providers-that-have-committed>.
- [29] Office of the National Coordinator for Health Information Technology. *ONC HTI-1 Final Rule (2023): Health Data, Technology, and Interoperability Certification Updates, Algorithm Transparency, and Information Sharing*. Dec. 2023. URL: <https://healthit.gov/regulations/hti-rules/hti-1-final-rule/>.
- [30] U.S. Food and Drug Administration. *Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products*. Jan. 2025. URL: <https://tinyurl.com/yvz5n24j>.

- [31] Kevin Kawchak. *Patient Instructions: Physical AI Oncology Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/patients>. Mar. 2026. DOI: 10.5281/zenodo.18810541. URL: <https://doi.org/10.5281/zenodo.18810541>.
- [32] Kevin Kawchak. *Unification Standard Level for Physical AI Oncology Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/unification/us1>. Feb. 2026. DOI: 10.5281/zenodo.18778219. URL: <https://doi.org/10.5281/zenodo.18778219>.
- [33] TARGET Investigators. “A Prospective, Multicenter Evaluation of Safety and Diagnostic Outcomes With Robotic-Assisted Bronchoscopy: Results of the Transbronchial Biopsy Assisted by Robot Guidance in the Evaluation of Tumors of the Lung (TARGET) Trial”. In: *CHEST* 168.2 (2025), pp. 539–555. DOI: 10.1016/j.chest.2025.04.022. URL: <https://doi.org/10.1016/j.chest.2025.04.022>.
- [34] U.S. Food and Drug Administration. *Informed Consent: Guidance for IRBs, Clinical Investigators, and Sponsors*. Aug. 2023. URL: <https://www.fda.gov/media/88915/download>.
- [35] Eugene Vorontsov, Alican Bozkurt, Adam Casson, et al. “A Foundation Model for Clinical-Grade Computational Pathology and Rare Cancers Detection”. In: *Nature Medicine* 30 (2024), pp. 2924–2935. DOI: 10.1038/s41591-024-03141-0. URL: <https://doi.org/10.1038/s41591-024-03141-0>.
- [36] Office for Human Research Protections. *Attachment C: Recommendations for Broad Consent Guidance, August 2017*. Aug. 2017. URL: <https://tinyurl.com/3fteucp5>.
- [37] United States Congress. *21st Century Cures Act, PL 114-255 (2016)*. 2016. URL: <https://www.congress.gov/bill/114th-congress/house-bill/34>.
- [38] U.S. Food and Drug Administration and Office for Human Research Protections. *Key Information and Facilitating Understanding in Informed Consent: Draft Guidance for Sponsors, Investigators, and Institutional Review Boards*. Mar. 2024. URL: <https://tinyurl.com/5ay93kyb>.
- [39] Kevin Kawchak. *Adaption of ICH E6(R3) for Physical AI Oncology Clinical Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/regulatory>. Mar. 2026. DOI: 10.5281/zenodo.18973368. URL: <https://doi.org/10.5281/zenodo.18973368>.
- [40] Kevin Kawchak. *Adaption of 21 CFR Part 50 for Physical AI Oncology Clinical Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/regulatory>. Mar. 2026. DOI: 10.5281/zenodo.19040707. URL: <https://doi.org/10.5281/zenodo.19040707>.

- [41] Assistant Secretary for Technology Policy and Office of the National Coordinator for Health Information Technology. *ONC's Cures Act Final Rule*. 2026. URL: <https://www.healthit.gov/regulations/cures-act-final-rule/>.
- [42] Nora Kearney, Lisa McCann, John Norrie, et al. "Evaluation of a Mobile Phone-Based, Advanced Symptom Management System (ASyMS) in the Management of Chemotherapy-Related Toxicity". In: *Supportive Care in Cancer* 17.4 (2009), pp. 437–444. DOI: 10.1007/s00520-008-0515-0. URL: <https://doi.org/10.1007/s00520-008-0515-0>.
- [43] Roma Maguire, Patricia A. Fox, Lisa McCann, et al. "The eSMART Study Protocol: A Randomised Controlled Trial to Evaluate Electronic Symptom Management Using the Advanced Symptom Management System (ASyMS) Remote Technology for Patients with Cancers". In: *BMJ Open* 7 (2017), e015016. DOI: 10.1136/bmjopen-2016-015016. URL: <https://doi.org/10.1136/bmjopen-2016-015016>.
- [44] Kate Absolom, Lorraine Warrington, Eleanor Hudson, et al. "Phase III Randomized Controlled Trial of eRAPID: eHealth Intervention During Chemotherapy". In: *Journal of Clinical Oncology* 39.7 (2021), pp. 734–747. DOI: 10.1200/JCO.20.02015. URL: <https://doi.org/10.1200/JCO.20.02015>.
- [45] Brian D. Gonzalez et al. "Using Bayesian Networks to Predict Urgent Care Visits in Patients Receiving Systemic Therapy for Non-Small Cell Lung Cancer". In: *JCO Clinical Cancer Informatics* 9.9 (2025), e2400315. DOI: 10.1200/CCI-24-00315. URL: <https://doi.org/10.1200/CCI-24-00315>.
- [46] U.S. Food and Drug Administration. *Digital Health Technologies (DHTs) for Drug Development*. 2026. URL: <https://www.fda.gov/science-research/science-and-research-special-topics/digital-health-technologies-dhts-drug-development>.
- [47] National Institutes of Health. *PAR-25-170: Digital Health Technology Derived Endpoints in Cancer Clinical Trials*. Nov. 2024. URL: <https://grants.nih.gov/grants/guide/pa-files/PAR-25-170.html>.
- [48] Kevin Kawchak. *Physical AI for Oncology Clinical Trials*. GitHub: <https://github.com/kevinkawchak/physical-ai-oncology-trials>. 2026. DOI: 10.5281/zenodo.18445179. URL: <https://doi.org/10.5281/zenodo.18445179>.
- [49] Anthropic. *Claude Opus 4.7 (1M Token Max Context)*. 2026. URL: <https://www.anthropic.com/claude/opus>.
- [50] Anthropic. *Claude Code*. 2026. URL: <https://claude.com/product/claude-code>.
- [51] Anthropic. *Claude Sonnet 4.6*. 2026. URL: <https://tinyurl.com/ycx4r7ue>.
- [52] OpenAI. *ChatGPT (Thinking, Extended Thinking, and Deep Research)*. 2026. URL: <https://chatgpt.com/>.
- [53] Google. *Google Gemini AI Overview*. 2026. URL: <https://gemini.google.com/>.

Acknowledgments

The author acknowledges Anthropic for providing access to Claude Code for the paper generations [50, 49, 51]; OpenAI for providing access to ChatGPT for detailed deep research summaries [52]; and Google Gemini AI overview for additional search assistance [53]. The author further acknowledges the authors of the bibliographic source material from which this paper draws, including the FDA, OHRP, ONC, HHS, CMS, the United States Congress, and the journal article authors cited herein.

Ethical disclosures

The author of the article declares no competing interests. The seven proposed bills (HR 9501 through HR 9507) are independent draft language and have not been reviewed by Congress, the FDA, OHRP, ONC, HHS, CMS, or any state legislature. Bill numbers and short titles are illustrative placeholders selected to avoid known active legislation under the 119th Congress; the legal-design intent of each section remains the substantive contribution.

Rights and permissions

This article is distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author(s) and source are properly credited, a link to the Creative Commons license is provided, and any modifications made are indicated. To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>.

Cite this article

Kawchak K. Patient Priority of Proposed U.S. Bills for Physical AI Oncology Clinical Trials. Zenodo. 2026; 10.5281/zenodo.20045457.